10.6.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on solving problems using the relationship between the measure of an inscribed angle and its intercepted arc?

Unit 10, Lesson 7: Solving Problems Involving Intersecting Chords

Benchmark

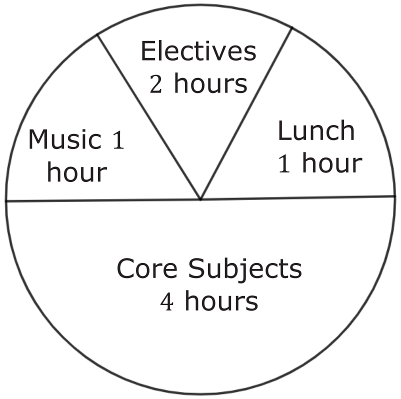
MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.

Learning Target

☐ I can solve problems involving chords of a circle and their related angles.

10.7.1 Warm-Up: Sectors

1. What do you notice about the circle graph?

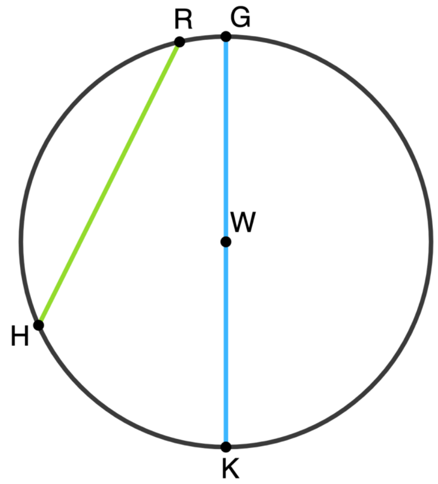


2. Identify a sector that is incorrectly sized.

3. Redraw the circle graph so that the sizes are proportional to the number of hours.

10.7.2 Exploration: Chords

Circle W is shown where points R, G, K, and H lie on the circle.



1. What similarities do you notice between $\overline{RH}$ and $\overline{GK}$?

2. What differences do you notice between $\overline{RH}$ and $\overline{GK}$?

A ***chord*** is a line segment that connects two points on the curve of a circle.

3. Complete the statements.

$\overline{RH}$ is
- ○ a diameter
- ○ a chord
- ○ both a diameter and a chord

of the circle.

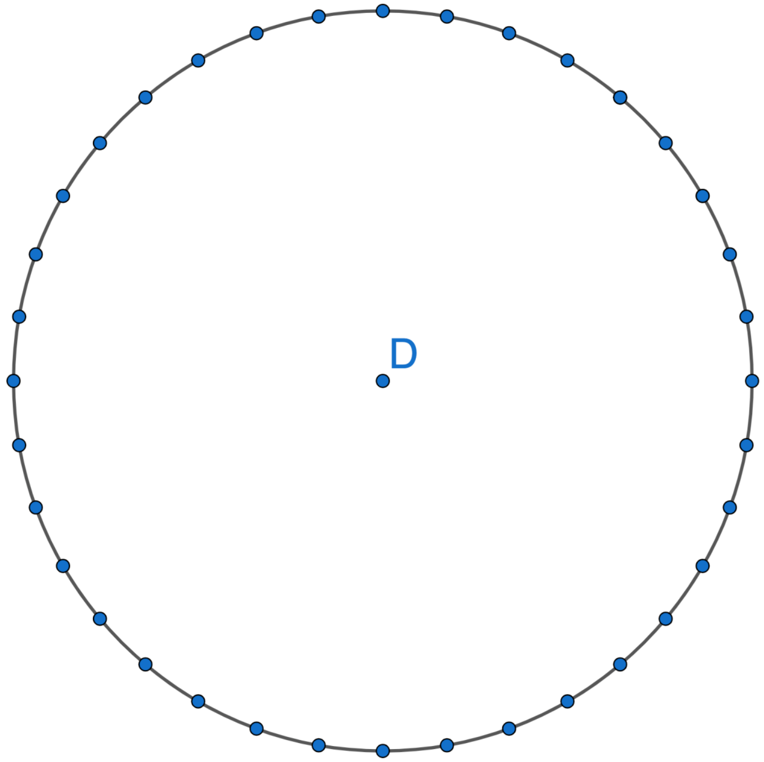
$\overline{GK}$ is
- ○ a diameter
- ○ a chord
- ○ both a diameter and a chord

of the circle.

10.7.3 Collaboration: Angles in Intersecting Chords

1. With your partner, complete the following using Circle D.

   a. Draw and label two chords that intersect inside the circle, making sure the endpoints fall on the points. When drawing the chords, do not draw the same chords as your partner and avoid drawing diameters.



b. Choose one of the angles to measure and label. Then, complete the table.

	Angle Measure	Measure of Arc it Intercepts	Measure of the Arc the Vertical Angle Intercepts
Your Angle			
Your Partner's Angle			

c. Discuss the following with your partner.

- What do you notice about the values in the table?

- What do you think the relationship is between the angle measure and the two intercepted arcs?

d. Write a formula that generalizes your discovery from the discussion you had in Part C.

e. Complete the statement.

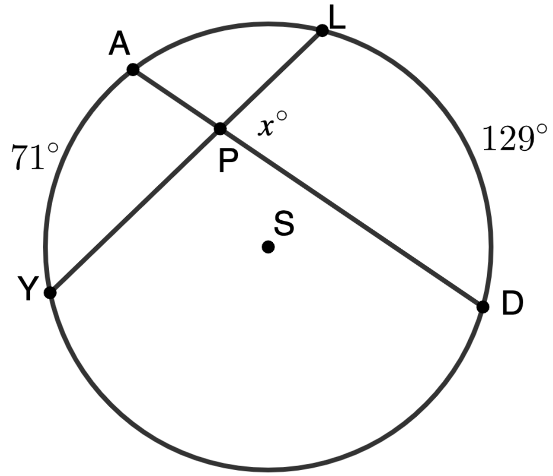
Angles of Intersecting Chords Theorem

The angle formed inside of a circle by two intersecting chords is
- ○ double
- ○ equal
- ○ half

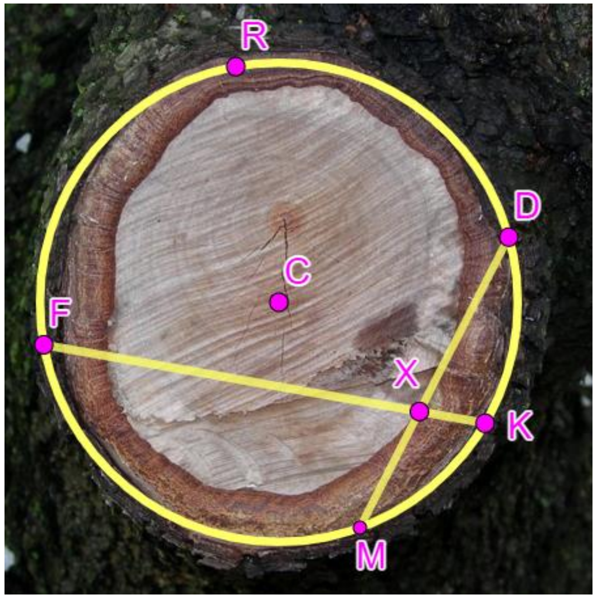
the sum of the measures of the arcs intercepted by the angle and its vertical angle.

10.7.4 Guided Practice: Solving Problems Involving Angles in Intersecting Chords

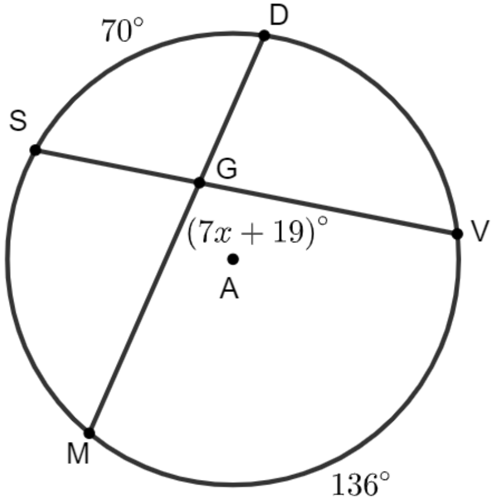
1. Circle S is shown, where chords $\overline{AD}$ and $\overline{YL}$ intersect at point P. Find the value of x for $\angle LPD$, given $m\,\widehat{AY} = 71°$ and $m\,\widehat{LD} = 129°$.



2. A tree is cut down and a circular tree stump is left. The top of the tree stump has two cracks that represent two chords, $\overline{FK}$ and $\overline{DM}$, intersecting at point X in Circle C. If $m\,\widehat{FRD} = 175°$ and $m\,\widehat{KM} = 39°$, what is $m\angle FXD$?

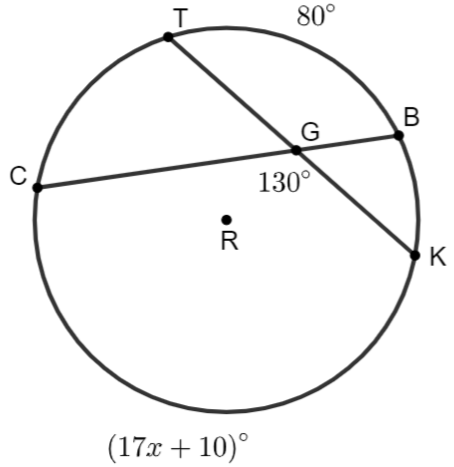


3. Circle A is shown, where chords $\overline{SV}$ and $\overline{MD}$ intersect at point G. Find the value of x, given $m\,\widehat{SD} = 70°$, $m\,\widehat{MV} = 136°$, and $m\angle MGV = (7x + 19)°$.



10.7.5 Your Turn!

1. Circle R is shown, where chords $\overline{TK}$ and $\overline{CB}$ intersect at point G. Find the value of x, given $m\,\widehat{TB} = 80°$, $m\,\widehat{CK} = (17x + 10)°$, and $m\angle CGK = 130°$.



10.7.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to solving problems involving chords.

1. I can define a chord of a circle.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. I can define the relationship of angles and arcs formed by the intersection of two chords.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

3. I can solve problems involving chords of a circle.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

Unit 10, Lesson 8: Solving Problems Involving Intersection of Secants

Benchmark

MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.

Learning Target

- [ ] I can solve problems involving secants drawn from a point external to the circle and their related angles.